Calibration-guided robust reallocation of emergency resources under evolving flood risk forecasts: a digital twin framework

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Abstract

In the first days of a major flood, authorities must place scarce rescue teams, boats and medical units while risk forecasts keep changing and roads close to floodwater and landslides. Existing allocation models describe forecast uncertainty with hand-tuned budgets, so the protection a plan offers is not tied to how reliable the forecasts are. This study links the two in a digital twin framework. Flood risk forecasts are recalibrated by temperature scaling and converted, through conformal prediction and an exposure model, into demand ranges with a chosen coverage $\mathbf{1-\alpha}$. A robust allocation model assigns resources with travel times that grow with predicted inundation, landslide-blocked routes removed and a cost for moving teams, and a trigger rule re-plans only when forecasts, risk classes or roads change enough. The framework is evaluated on hindcasts of the August 2018 Kerala floods and, as an independent test, the 2019 floods, against seven baselines. Temperature scaling reduced calibration error on 2019 but not on 2018, where forecasts were under-confident. In 2018 the framework covered 76.5% of modelled demand, against 55.6–65.6% for forecast-based, fixed-budget robust and learned policies, with 20 re-plans instead of 96. Two-stage stochastic programming covered more (85.2%) but re-planned every period with 74% more relocations; at $\boldsymbol{\alpha = 0.05}$ the framework reached 84.0% with 41 re-plans. Under the severe resource scarcity of 2019, all methods covered 27–30% of demand. The framework gives authorities one interpretable reliability setting and a stated reason for each redeployment.

Keywords: Flood emergency response, Forecast calibration, Conformal prediction, Robust optimization, Digital twin, Kerala floods

1 Introduction

Floods are among the most damaging recurrent hazards in India, and their impacts concentrate in districts where exposure and vulnerability are high (Saharia et al, 2025; Jain et al, 2025). In the first days of a major flood, where scarce rescue resources are placed decides how many trapped and displaced people are reached in time. The August 2018 Kerala floods killed 433 people, affected 5.4 million and displaced 1.4 million, and water was released from 37 dams (Government of Kerala and United Nations, 2018). About 341 landslides occurred during the same event, many of them in the hill districts of Idukki and Wayanad, cutting the roads on which rescue teams depended (Government of Kerala and United Nations, 2018). The National Disaster Response Force (NDRF) made its largest deployment in a single state, sending 58 teams across ten severely affected districts (National Disaster Response Force, 2018a). The difficulty was not only the number of teams but where they were placed as the flood spread, as dam releases changed river levels and as roads became impassable.

Forecasting has advanced faster than the decisions that depend on it. Deep learning models now map flood susceptibility at regional scale (Rachit et al, 2026), coupled hydrologic–hydraulic systems forecast inundation depth with 48-hour lead times (Rao et al, 2024), and conformal methods are being extended to give reliable intervals for extreme events (Pasche et al, 2026). Yet a forecast helps responders only when it is turned into an allocation that still holds up if the forecast is wrong. Model reliability also changes over time: a model that performs well within its calibration period can lose accuracy in later periods (Xiao et al, 2026). Decision models therefore need to know how far each forecast can be trusted, and need to adapt as forecasts are updated.

The step from hazard to decision is where the hazard literature is thinnest. A recent systematic review of 80 studies on landslides affecting roads found that 90% assessed susceptibility or hazard, but only 21% completed the chain from hazard to consequence to risk, only 9% modelled network connectivity and only 10% quantified indirect losses (Vega et al, 2026). Studies that do model emergency logistics usually take forecasts as given, with uncertainty budgets chosen by the modeller (Caunhye et al, 2012; Zhang et al, 2023). The protection a plan offers is therefore not tied to the measured reliability of the forecasts that drive it, and re-planning at every period can move teams back and forth as forecasts fluctuate.

This study builds a decision framework for flood response that closes that link. Its contribution is not a new flood prediction model: it shows how the reliability of whatever forecasts are available can be measured, improved and carried into allocation decisions. It recalibrates the forecast probabilities, converts them into demand ranges with a stated reliability, allocates resources so that the plan holds across those ranges on a road network that degrades with flood depth and landslide blockage, and re-plans only when the situation has changed enough to justify moving teams. The framework is tested on the 2018 Kerala floods and, as an independent check, on the August 2019 Kerala floods.

The study addresses three research questions:

RQ1. Do demand ranges built from calibrated flood risk forecasts reduce worst-case unmet demand compared with fixed or hand-tuned ranges, at similar operational cost?

RQ2. Can a forecast-triggered re-planning rule match the coverage of re-planning at every period with fewer re-plans and team relocations?

RQ3. Does linking travel times to predicted inundation depth and observed landslide blockages reduce failed dispatches on the roads that actually closed?

The contributions are as follows. (1) A calibration-to-decision chain: forecast probabilities are recalibrated by temperature scaling, and district-level conformal prediction sets of flood risk classes into demand ranges with a coverage interpretation, so that a single setting α controls how cautious the plan is. (2) A robust allocation model with inundation-dependent travel times, landslide road closures and an explicit cost for relocating teams, with two supporting properties: its robust form is exact and linear, and calibrated forecasts give a probabilistic bound on unmet demand. (3) A forecast-triggered re-planning rule that gives authorities a stated reason for each redeployment. (4) A hindcast evaluation on the 2018 Kerala floods, with the 2019 floods held out, against seven baselines including uncalibrated and calibrated forecast-based allocation, with ablations and stress scenarios, and recommendations for disaster management authorities.

Section 2 surveys the literature and Section 3 states the problem and the research gap. Section 4 presents the novel components and the proposed algorithm. Section 5 describes the study area, data and preprocessing, and Section 6 the methodology. Section 7 defines the evaluation metrics, Section 8 reports and discusses the results, and Section 9 concludes.

2 Literature survey

2.1 Flood risk forecasting and its reliability

Flood risk models in India now operate at several scales. A district-level flood severity index combines hazard, exposure and vulnerability for national flood management (Saharia et al, 2025), and analysis of flood risk attributes has been used to identify where flood policy can improve (Jain et al, 2025). At regional scale, CNN–U-Net segmentation has produced rapid flood susceptibility maps for the Indian Himalayan region (Rachit et al, 2026), and integrated hydrologic–hydraulic models provide real-time inundation forecasts with 48-hour lead times in eastern India (Rao et al, 2024). For Kerala 2018, Sentinel-1 synthetic aperture radar (SAR) flood maps have been produced with about 94% overall accuracy (Tiwari et al, 2020), and the geomorphic processes of the event have been linked to monsoon circulation and land use change (Chothodi et al, 2024). A common limitation is that model reliability is reported as an average accuracy, and that reliability can drift as conditions change between calibration and use (Xiao et al, 2026).

2.2 Calibrated uncertainty for hazard forecasts

Probabilistic forecasts are often miscalibrated: a predicted probability of 0.8 is not realized 80% of the time. Post-hoc calibration such as temperature scaling (Guo et al, 2017) corrects this and is assessed with the expected calibration error (ECE), Brier score and negative log-likelihood (NLL). Conformal prediction constructs prediction sets that contain the true outcome with probability at least $1-\alpha$ under exchangeability, without distributional assumptions (Vovk et al, 2005; Angelopoulos and Bates, 2023). Pasche et al (2026) combined extreme value statistics with conformal prediction to obtain reliable intervals for rare, high-impact events such as floods. Johnstone and Cox (2021) showed that conformal prediction regions can serve as robust-optimization uncertainty sets with finite-sample guarantees, illustrated on simulated examples. Their use for sizing the uncertainty in disaster response decisions has, to our knowledge, not been explored.

2.3 Emergency resource allocation under uncertainty

Optimization models for emergency logistics are reviewed by Caunhye et al (2012). Robust optimization protects a plan against every outcome in an uncertainty set (Ben-Tal et al, 2009; Bertsimas and Sim, 2004); Ben-Tal et al (2011) applied an affinely adjustable robust model to emergency logistics with time-varying demand, and Avishan et al (2023) used adjustable robust optimization for relief distribution when damaged roads make travel times uncertain. For floods, Manopiniwes and Irohara (2021) formulated a multi-period location–routing model for temporary depots with multimodal transport. Rolling-horizon models re-solve as demand is revealed, for example in a hybrid robust–fuzzy model for earthquake relief (Attar et al, 2026), and Zhang et al (2023) adjusted the uncertainty budget over a rolling horizon to reduce conservativeness as an epidemic evolved. Only about one in ten humanitarian logistics optimization studies involved practitioners, and model objectives often differed from practitioners' priorities (Rodríguez-Espíndola et al, 2023), which argues for models whose settings authorities can interpret.

2.4 Road network disruption by floods and landslides

Floodwater slows and eventually stops road vehicles. Pregnolato et al (2017) derived a depth–disruption function relating vehicle speed to standing water depth. He et al (2026) combined flood hazard curves with vehicle fragility curves and found that flood exposure can restrict ambulance access to high-risk areas. Landslides add a second source of closure: Postance et al (2017) assessed landslide disruptions on a national road network, Zhang et al (2020) linked landslide likelihood to transport network vulnerability, and Zhang et al (2021) quantified indirect economic losses of landslides along highways. Globally, road risk from landslides is projected to rise under climate change (Duan et al, 2025). These studies quantify disruption but rarely feed it into the allocation of response resources (Vega et al, 2026).

2.5 Digital twins and learning-based dispatch

Digital twins couple live observations with simulation models of a hazard and the systems it affects. Liu et al (2026) combined a digital twin, an optimization model and a deep Q-network for urban emergency distribution during the Zhengzhou "7·20" rainstorm. Reviews of digital twin cities for flood evacuation (Prihatmanto et al, 2025), of digital twins in safety and emergency management (Zio and Miqueles, 2024), and of digital twin-based decision support for natural disasters (Inyang and Taghikhah, 2025) identify data integration, model accuracy and limited treatment of uncertainty as open problems. Reinforcement learning has been used to dispatch rescue teams during floods (Yan et al, 2020) and to assign rescue helicopters (Shen et al, 2023). These methods respond quickly once trained, but offer no coverage guarantee and are hard for authorities to audit.

2.6 Comparison with closely related studies

Table 1 compares this study with the closest work.

Table 1 Comparison with closely related studies

Study	Dynamic re-planning	Sets from calibrated forecasts	Coverage guarantee	Inundation-dependent travel	Landslide road closures	Relocation cost or trigger	Real flood case
Liu et al (2026)	Yes	No	No	Partly	No	No	Yes
Attar et al (2026)	Yes	No	No	No	No	No	No
Zhang et al (2023)	Yes	No	No	No	No	No	No
Yan et al (2020)	Yes	No	No	Yes	No	No	Yes
Avishan et al (2023)	Yes	No	No	No	Partly	No	No
Manopiniwes and Irohara (2021)	Yes	No	No	No	No	No	Yes
Johnstone and Cox (2021)	No	Yes	Yes	No	No	No	No
This study	Yes	Yes	Yes	Yes	Yes	Yes	Yes

3 Problem statement and research gap

3.1 Problem statement

A state disaster management authority holds a fixed stock of rescue teams, boats and medical units at staging bases. Over a flood event divided into decision periods $t = 1, \ldots, T$, it must decide how many units of each resource to send from each base to each affected unit (district or taluk). At each period it receives updated forecasts of the flood risk class of every unit and of inundation depth along roads, together with reports of roads closed by landslides. The true number of people needing assistance in each unit is unknown when the decision is made. The authority wants to (i) minimize unmet demand, weighted towards vulnerable units, including in the worst case across plausible forecast errors; (ii) reach people quickly; (iii) avoid sending resources along roads that are flooded beyond their capability or blocked; and (iv) avoid moving teams back and forth when forecasts change only slightly.

3.2 Research gap

The literature in Section 2 leaves four gaps.

G1. **Uncertainty in allocation is not tied to forecast reliability.** Robust and rolling-horizon allocation models size their uncertainty sets by hand or heuristically (Ben-Tal et al, 2011; Zhang et al, 2023; Attar et al, 2026), so a plan's protection has no stated link to how often the forecasts that drive it are wrong.

G2. **Hazard studies rarely reach the decision stage.** Most studies stop at susceptibility or hazard; few model network consequences or indirect losses, and fewer still turn them into response decisions (Vega et al, 2026).

G3. **Road disruption is treated separately from allocation.** Depth–disruption and landslide–network studies quantify which roads fail (Pregnolato et al, 2017; Postance et al, 2017; He et al, 2026), but allocation models rarely use predicted depth or landslide blockage to decide which resource type can reach a unit.

G4. **Re-planning has no stated rule.** Rolling-horizon and learned dispatch methods re-plan every period (Attar et al, 2026; Yan et al, 2020; Liu et al, 2026), which can move teams back and forth and gives authorities no reason for each redeployment.

RQ1 addresses G1 and G2, RQ2 addresses G4 and RQ3 addresses G3.

4 Novelty and proposed algorithm

4.1 Novel elements

The framework introduces five elements that, to our knowledge, have not been combined for flood emergency response. (N0) *Calibration before decision*: forecast probabilities are recalibrated (temperature scaling) and their reliability measured before they drive allocation (Sect. 6.1). (N1) *Calibrated demand ranges*: conformal prediction sets of flood risk classes are mapped through exposure and vulnerability into demand ranges with coverage at least $1 - \alpha$ (Sect. 6.2). (N2) *Data-driven uncertainty budget*: the number of units allowed to reach their upper demand together is set from calibration data rather than by hand (Definition 1). (N3) *Hazard-aware reachability*: travel times grow with predicted inundation depth, and routes deeper than a resource can cross or blocked by mapped landslides are excluded (Eq. 4). (N4) *Forecast-triggered re-planning*: the plan is revised only on a material forecast shift, a risk-class escalation or a route failure (Sect. 6.5). We call the combined method calibration-guided robust reallocation (CGRR). Novelty is claimed for the combination and its application to flood response; conformal sets for robust optimization exist in general (Johnstone and Cox, 2021).

4.2 Framework overview

The framework repeats four stages at each decision period (Fig. 1). (i) The forecasting layer issues risk class probabilities, which are recalibrated and turned into conformal prediction sets, together with predicted inundation depths; the road layer reports landslide closures. (ii) An exposure model converts the conformal sets into unit demand ranges and a budgeted uncertainty set. (iii) A trigger rule decides whether the current

plan should be revised. (iv) If so, a robust allocation model computes a new plan, which the digital twin then executes against the realized flood.

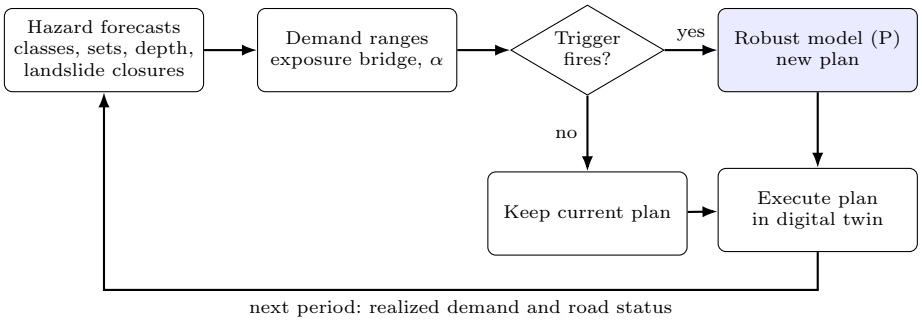


Fig. 1 The framework loop at each decision period. The robust model is solved only when the trigger fires; otherwise the current plan stays in force

4.3 Proposed algorithm

Algorithm 1 gives the complete procedure for one event. Equations are defined in Section 6.

Algorithm 1 Calibration-guided robust reallocation (CGRR) over one flood event

Require: calibration data; level α; trigger threshold η; stocks $S_{r,j}$; road network

fit split conformal predictor on calibration periods; set Γ from calibration data (Def. 1)
$\bar{x} \leftarrow$ undefined
for $t = 1, \ldots, T$ **do**
    obtain forecasts $(\hat{p}_t, C_t, \hat{h}_t)$ and landslide closures L_t
    compute $d_t^L, \hat{d}_t, d_t^U$ by (2) and $\mathcal{U}_t(\alpha)$ by (3)
    compute $\tau_{j,i,t}$ and passability by (4), removing routes in L_t
    if $\bar{x}$ undefined **or** $\Delta_t > \eta$ **or** an upper class of C_t rises **or** a used route is impassable **then**
        solve (P) with reference plan $\bar{x}$; set $\bar{x} \leftarrow x_t^*$, $\tau \leftarrow t$
    end if
    execute $\bar{x}$ in the digital twin; record realized demand, service and travel times
end for

5 Dataset and data preprocessing

5.1 Study area: the August 2018 Kerala floods

Kerala, on India's south-west coast, experienced its most severe flooding in decades in August 2018. Intense rainfall from the night of 8–9 August, together with water

released from 37 dams, flooded river basins across all 14 districts and triggered about 341 landslides (Government of Kerala and United Nations, 2018; National Disaster Response Force, 2018a). NDRF deployed 58 teams, concentrated in Pathanamthitta, Idukki, Alappuzha, Wayanad, Kottayam, Ernakulam, Thrissur, Kozhikode, Thiruvananthapuram and Malappuram districts; its teams rescued 535 people and evacuated 24,690 (National Disaster Response Force, 2018a). In total, 1.4 million people were displaced and 1,259 of the state's 1,664 villages were affected (Government of Kerala and United Nations, 2018).

The event suits this study for three reasons. Demand shifted across districts within days as dam releases moved floodwater downstream. Rescue resources were of several types, with different abilities to cross deep water. And flood extent, landslide, population and road data are publicly available, which allows a hindcast. The August 2019 Kerala floods, which again combined riverine flooding with landslides in the hill districts, including the Puthumala (Wayanad) and Kavalappara (Malappuram) landslides of 8 August 2019 (National Disaster Response Force, 2019), serve as an independent test event that is never used for calibration or tuning.

5.2 Data sources

The decision units are the 14 districts of Kerala, $|I| = 14$. The 2018 hindcast covers 8–31 August 2018 in 96 six-hour decision periods, and the 2019 test covers 5–25 August 2019 in 84 periods. Resources are rescue teams, boats and medical units staged at four hubs: Thrissur, where NDRF has its regional response centre in Kerala, and Ernakulam, Thiruvananthapuram and Kozhikode, the three cities served by the international airports through which airlifted teams arrived. Because district headquarters double as hubs, the hubs are the district origins of the travel-time matrix. For 2018 the stock is the 55 NDRF teams deployed simultaneously at the peak on 18 August (National Disaster Response Force, 2018b), rather than the cumulative total of 58 (National Disaster Response Force, 2018a); for 2019 it is the 9 teams deployed for the landslides (National Disaster Response Force, 2019). Teams are split evenly across the hubs (14, 14, 14 and 13 in 2018; 3, 2, 2 and 2 in 2019). The split, the boat stock (two per team: 110 in 2018, 18 in 2019) and the medical units (four per hub in 2018, two in 2019) are assumptions, not records; stress scenario S2 tests 25% fewer resources. Table 2 lists the data.

5.3 Data preprocessing

All methods compared in Section 8 use the same preprocessed inputs, so that differences in results come from the allocation method alone.

5.3.1 Flood extent

Flood extent is mapped from Sentinel-1 ground-range-detected scenes in vertical–vertical (VV) polarization, following Tiwari et al (2020), in Google Earth Engine. Each flood-date mosaic is speckle-filtered (50 m focal median) and compared with a dry-season reference (median of April–May scenes from the same orbit direction). A pixel is flooded if VV backscatter is below -15 dB and has dropped by at least 3 dB from

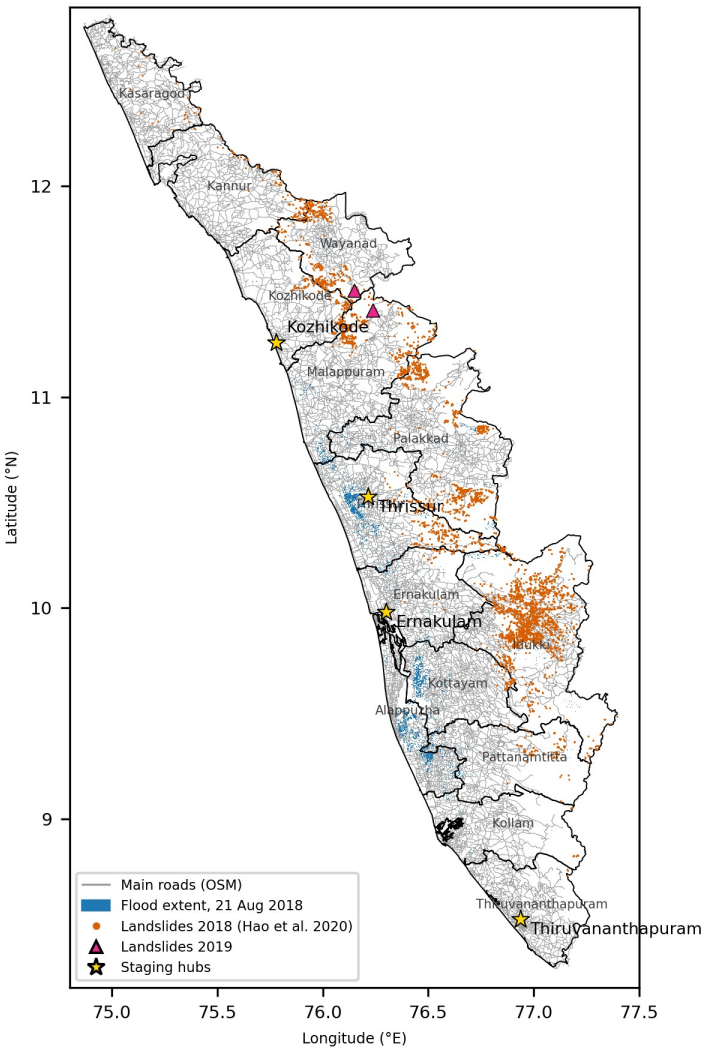


Fig. 2 Study area: the 14 districts of Kerala (decision units), the main road network (OpenStreetMap, motorway to tertiary), Sentinel-1 flood extent on 21 August 2018, the 4,728 landslides of the 2018 inventory (Hao et al, 2020), the two 2019 landslide sites and the four staging hubs

the reference. Permanent water (JRC Global Surface Water, water at least 10 months a year) and slopes steeper than 5° (SRTM) are masked. Usable scenes cover 9, 14, 15, 21, 26 and 27 August 2018 and 10, 21 and 22 August 2019. Each scene is assigned to the period in which it was acquired, and between acquisitions the flood extent is held at the last observation. A district imaged over less than half of its area on a given date is treated as unobserved on that date and keeps its last observed state; without this rule, districts that Sentinel-1 did not image on 14–15 August 2018, at the peak of

Table 2 Data used in the study

Data	Use in the model	Source
Risk class probabilities	Calibration, conformal sets, demand ranges (Sects. 6.1–6.2)	Rainfall-based flood risk model (ridge regression on antecedent rainfall and a static low-lying-area index, 24-h lead), issued every 6 h for each district
Predicted inundation depth	Travel time and passability	Proxy: 1.5 m $\times P$(high or severe), because the forecasting model gives no depth
Observed flood extent	Ground truth for forecasts; realized demand; flooded roads	Sentinel-1 SAR change detection in Google Earth Engine, following Tiwari et al (2020)
Rainfall	Forecasting-model input	NASA GPM IMERG V07 (Huffman et al, 2023)
Landslide locations	Road closures	2018: inventory of 4,728 landslides (Hao et al, 2020); 2019: the Puthumala and Kavalappara landslides (National Disaster Response Force, 2019)
Relief-camp occupancy	Independent check of realized demand	District-wise camps and inmates, 2018 (Government of Kerala, 2018); no district-wise record found for 2019
Population	Exposed population $P_{i,t}$, P_i	WorldPop 100 m (WorldPop, 2018)
Road network	Normal travel times $\tau^0_{j,i}$; flooded and blocked segments	OpenStreetMap (OpenStreetMap contributors, 2026)
Resources and bases	$S_{r,j}$, κ_r	NDRF situation reports (National Disaster Response Force, 2018a,b, 2019); boat and medical stocks assumed
Need share and damage fraction	Realized demand; δ_k	Assumed need share (Sect. 5); δ_k estimated on the calibration periods

the flood, would appear dry. Before the first scene of each event (8 August 2018; 5–9 August 2019) observed exposure is zero.

5.3.2 Exposed population

WorldPop 100 m population for the event year (WorldPop, 2018) is overlaid on the flood extent of each scene to give the exposed population $P_{i,t}$ of each district; total district population gives P_i.

5.3.3 Road network and travel times

The OpenStreetMap road network (OpenStreetMap contributors, 2026) of Kerala (motorway to tertiary roads) is converted to a routable graph with OSMnx. Normal travel times $\tau^0_{j,i}$ are shortest-path times between a representative interior point of each district, using road-class speeds. A road segment is flooded on a scene date if any point sampled every 100 m along it lies in the flood mask; realized road and boat travel times per date are recomputed on the network with flooded and blocked segments removed (boats may use flooded segments but not landslide-blocked ones). Because the forecasting model gives no depth, the predicted depth on a route is approximated as $\hat{h}_{j,i,t} = 1.5\,\mathrm{m} \times (\hat{p}_{i,t,\mathrm{high}} + \hat{p}_{i,t,\mathrm{severe}})$.

5.3.4 Landslide road closures

Mapped landslides within 50 m of a road segment close that segment. The 2018 inventory (Hao et al, 2020) does not give a reliable date for every landslide, so all are treated as blocking from 15 August 2018, the onset of the main landslide episode, to the end of the event; this closes 317 segments of the main-road network. The 2019 sites block from 8 August 2019, but neither lies within 50 m of a main-road segment, so 2019 has no landslide closures in the model. Reopening dates are not recorded, so closures last until the end of the event. Routes that cross a closed segment are recomputed or marked impassable.

5.3.5 Vulnerability

No district vulnerability index was used in this study: $w_i = v_i = 1$ for all districts, so the equity floor (9) is inactive and all people in need count equally. The framework accepts any index, such as the District Flood Severity Index (Saharia et al, 2025), without change.

5.3.6 Realized demand and its check

Realized demand in each district and period is the observed exposed population multiplied by a need share, the fraction of exposed people needing rescue, evacuation or medical help. The need share is drawn for each district and simulation run from a Beta$(2, 98)$ distribution (mean 2%), an assumption tested in Sect. 8 with means of 1%, 2% and 5%. The damage fractions δ_k used on the forecast side are estimated on the calibration periods as the mean exposed fraction of each observed class multiplied by the mean need share. As a check that does not depend on the model, the peak exposed population of each district in 2018 was compared with district-wise relief-camp inmates (Government of Kerala, 2018): the Spearman rank correlation is $\rho = 0.60$ ($p = 0.02$, $n = 14$). Camp counts are cumulative over August and include people from unflooded areas, so only the ranking is compared. The largest mismatches are Ernakulam and Pathanamthitta, where camp numbers were high but mapped exposure was low, consistent with SAR under-detection of flooding in urban and forested areas.

5.3.7 Calibration and test split

Splits are chronological, never random, because nearby units and consecutive periods are correlated. The forecasting model is trained on June–September 2016 and June–July 2017 (160 district-scene observations). The temperature, the conformal predictor, the damage fractions and Γ are fitted on August–September 2017 and June–July 2018 (136 observations on 19 Sentinel-1 dates; 62 low, 49 medium, 18 high and 7 severe). The 2018 event is the main test and the 2019 event a held-out test. Neither is used for calibration or tuning: α, η, β, λ and μ were fixed before the experiments (Online Resource 1) and are varied only in the sensitivity analysis.

6 Methodology

6.1 Forecast calibration

Let $\tilde{p}_{i,t,k}$ be the raw forecast probability of risk class k for unit i at time t. Temperature scaling (Guo et al, 2017) rescales the log-probabilities with a single parameter $T > 0$,

$$\hat{p}_{i,t,k} = \frac{\exp\big(\log \tilde{p}_{i,t,k}/T\big)}{\sum_{l \in K} \exp\big(\log \tilde{p}_{i,t,l}/T\big)}, \tag{1}$$

where T minimizes the negative log-likelihood of the observed classes over the calibration periods. $T > 1$ softens over-confident forecasts and leaves the predicted class unchanged. All methods except the uncalibrated baseline (FU) use $\hat{p}$.

6.2 From conformal prediction sets to demand ranges

For unit i at time t, the forecasting model gives class probabilities $\hat{p}_{i,t,k}$ over risk classes $k \in K = \{\text{low}, \text{medium}, \text{high}, \text{severe}\}$ and a conformal prediction set $C_{i,t} \subseteq K$ with coverage at least $1-\alpha$. Each class has a damage fraction δ_k, the share of exposed people needing assistance. With exposed population P_i and vulnerability multiplier v_i,

$$d^L_{i,t} = v_i P_i \min_{k \in C_{i,t}} \delta_k, \qquad d^U_{i,t} = v_i P_i \max_{k \in C_{i,t}} \delta_k, \qquad \hat{d}_{i,t} = v_i P_i \sum_{k \in K} \hat{p}_{i,t,k}\, \delta_k. \tag{2}$$

Definition 1 (Calibrated budgeted uncertainty set)**.** For a budget $\Gamma_t \in \{0, 1, \ldots, |I|\}$,

$$\mathcal{U}_t(\alpha) = \Big\{ d : d_{i,t} = \hat{d}_{i,t} + z_i\,(d^U_{i,t} - \hat{d}_{i,t}),\ 0 \le z_i \le 1,\ \sum_i z_i \le \Gamma_t \Big\}. \tag{3}$$

Γ_t limits how many units can reach their upper bound together. It is set from the calibration data as the 95th percentile, over calibration periods, of the number of units whose realized damage fraction exceeded the point estimate. Smaller α widens the ranges and makes plans more cautious.

6.3 Robust reallocation model

Let I be the units, J the staging bases and R the resource types. Resource r serves κ_r people per period, base j holds $S_{r,j}$ units of it, and w_i is the vulnerability weight of unit i. The normal travel time is $\tau^0_{j,i}$, the predicted maximum inundation depth on the route is $\hat{h}_{j,i,t}$, and resource r can cross water up to depth $h^{\max}_r$. The decision $x_{r,j,i,t} \in \mathbb{Z}_+$ is the number of units of r sent from j to i; $\bar{x}_{t-1}$ is the plan in force.

Travel times grow with predicted inundation, a linear approximation of the depth–disruption relation of Pregnolato et al (2017). Routes that are too deep, too slow or blocked by a landslide (L_t is the set of blocked base–unit routes) are excluded:

$$\tau_{j,i,t} = \tau^0_{j,i}\,\big(1 + \beta\,\hat{h}_{j,i,t}\big), \qquad x_{r,j,i,t} = 0 \ \text{ if } \ \hat{h}_{j,i,t} > h^{\max}_r,\ \tau_{j,i,t} > \tau^{\max} \text{ or } (j,i) \in L_t. \tag{4}$$

Boats are given a large $h_r^{\max}$ but cannot use routes blocked by landslides on hill roads. In this study $\hat{h}_{j,i,t}$ is the depth proxy of Sect. 5. With coverage $\mathrm{cov}_{i,t} = \sum_r \sum_j \kappa_r x_{r,j,i,t}$, model (P) is

$$\begin{aligned}
\min\ & \sum_i w_i u_{i,t} + \Gamma_t \theta_t + \sum_i \pi_{i,t} + \lambda \sum_{r,j,i} \tau_{j,i,t}\, x_{r,j,i,t} \\
& \quad + \mu \sum_{r,j,i} \left(y^{+}_{r,j,i,t} + y^{-}_{r,j,i,t} \right) + M \sum_{i \in I^{\mathrm{v}}} w_i e_{i,t} && (5) \\
\text{s.t. } & u_{i,t} \geq \hat{d}_{i,t} - \mathrm{cov}_{i,t}, \quad a_{i,t} \geq d^{U}_{i,t} - \mathrm{cov}_{i,t} && \forall i, \quad (6) \\
& \theta_t + \pi_{i,t} \geq w_i \left(a_{i,t} - u_{i,t} \right) && \forall i, \quad (7) \\
& \textstyle\sum_i x_{r,j,i,t} \leq S_{r,j} && \forall r, j, \quad (8) \\
& \mathrm{cov}_{i,t} + e_{i,t} \geq \rho\, d^{L}_{i,t} && \forall i \in I^{\mathrm{v}}, \quad (9) \\
& x_{r,j,i,t} - \bar{x}_{r,j,i,t-1} = y^{+}_{r,j,i,t} - y^{-}_{r,j,i,t} && \forall r, j, i, \quad (10) \\
& x \in \mathbb{Z}_{+},\ u, a, y^{\pm}, e, \theta, \pi \geq 0.
\end{aligned}$$

Here u is nominal unmet demand, a is unmet demand at the upper bound, θ and π are dual variables of the robust counterpart, and λ and μ weight response time and relocation. Constraint (9) is an equity floor for the most vulnerable units I^{v}; it is soft, with penalty M, so the model stays feasible when flooded or blocked roads make a vulnerable unit unreachable. Model (P) is solved one period at a time; the relocation term links consecutive periods.

6.4 Properties of the model

Proposition 1. *For fixed coverage and integer Γ_t, the minimum of $\sum_i w_i u_{i,t} + \Gamma_t \theta_t + \sum_i \pi_{i,t}$ subject to (6)–(7) and non-negativity equals the worst-case weighted unmet demand over $\mathcal{U}_t(\alpha)$.*

Proof. See Appendix A. □

In plain terms, Proposition 1 means the linear model (P) protects against the worst combination of up to Γ_t units reaching their upper demand, with no approximation, so standard solvers can be used.

Proposition 2. *Assume (A1) calibration and test units are exchangeable, so that $P(k_i^{\star} \in C_i) \geq 1 - \alpha$ for the true class $k_i^{\star}$, and (A2) realized demand satisfies $d_i \leq v_i P_i \delta_{k_i^{\star}}$. Then for any plan x, the realized shortfall s_i of each unit satisfies $P\big(s_i \leq a_i(x)\big) \geq 1 - \alpha$, and*

$$\mathbb{E}\Big[\sum_i w_i \big(s_i - a_i(x) \big)^{+} \Big] \ \leq\ \alpha \sum_i w_i\, v_i P_i \big(\delta_{\max} - \delta_{\min} \big), \qquad (11)$$

where $\delta_{\max}$ and $\delta_{\min}$ are the largest and smallest damage fractions.

Proof. If $k_i^{\star} \in C_i$, then $d_i \leq v_i P_i \delta_{k_i^{\star}} \leq d_i^{U}$, so $s_i = (d_i - \mathrm{cov}_i)^{+} \leq (d_i^{U} - \mathrm{cov}_i)^{+} = a_i(x)$. Otherwise $(s_i - a_i(x))^{+} \leq (d_i - d_i^{U})^{+} \leq v_i P_i (\delta_{\max} - \delta_{\min})$, a constant. Hence

$(s_i - a_i(x))^+ \le \mathbf{1}\{k_i^\star \notin C_i\}\, v_i P_i(\delta_{\max} - \delta_{\min})$, and the event $k_i^\star \notin C_i$ has probability at most α by (A1). Taking expectations and summing gives (11). By linearity, the expected number of units whose shortfall exceeds $a_i(x)$ is at most $\alpha|I|$. □

Remark 1. Split conformal prediction guarantees marginal coverage under exchangeability. Flood periods are serially dependent, so (A1) may hold only approximately; Section 8 therefore reports the empirical coverage on both test events, and stress scenario S3 measures what happens when coverage falls below its target. Class-conditional (Mondrian) conformal prediction can be used if guarantees are needed per risk class.

6.5 Forecast-triggered reallocation

Re-solving (P) at every period can move teams back and forth as forecasts fluctuate. The plan in force since time τ is therefore revised only if at least one of three conditions holds. (1) Forecast shift: the weighted change in point demand, relative to the larger of the reference demand and the total response capacity per period $C = \sum_{r,j} \kappa_r S_{r,j}$, exceeds a threshold η,

$$\Delta_t = \frac{\sum_i w_i \, |\hat{d}_{i,t} - \hat{d}_{i,\tau}|}{\max\left(\sum_i w_i \, \hat{d}_{i,\tau},\; C\right)} > \eta. \tag{12}$$

The capacity floor stops near-zero demand early and late in the event from triggering re-planning. (2) Risk-class escalation: the upper class of $C_{i,t}$ rises for any unit. (3) Route failure: a route used by the current plan becomes impassable through flooding or a landslide. The rule plays the same role as tiered warning levels in operational early warning, where action is taken only when a defined level is crossed (Metodiev et al, 2026); here every redeployment comes with the condition that caused it.

6.6 Use of AI tools

Claude (Anthropic), a large language model, was used to help write and debug the analysis code, run the Colab notebooks and draft and edit parts of the manuscript text. The authors specified the study design, checked all code, data and numbers against the source files and run logs, and take full responsibility for the content.

6.7 Hindcast simulator

The digital twin replays each event period by period. Plans are made only from forecasts available at decision time and are then scored against realized demand and realized road closures; no method sees realized values when planning. Uncertainty in realized demand is represented by drawing the need share of each district for each run from Beta$(2, 98)$; each configuration is run 200 times with the same draws for every method, so that comparisons are paired. Optimization models are solved with HiGHS (Huangfu and Hall, 2018) via PuLP 2.9 in Python 3.13 on a Google Colab CPU runtime; the full configuration, input checksums and run log are listed in Online Resource 1 and in the code repository.

6.8 Baselines and ablations

Table 3 lists the compared methods. All use the same inputs, decision periods and road closures. FU and FC solve the nominal version of (P) ($\Gamma_t = 0$) on the raw and calibrated point forecasts; comparing FU with FC isolates the value of calibration, and FC with CGRR the value of calibrated uncertainty sets and triggered re-planning. B5 is a greedy dispatcher whose three score weights are fitted by random search (60 candidates) on demand scenarios simulated from the forecasts of the planning periods; it never sees realized demand or road status. Ablation A1 replaces the conformal sets with fixed-width ($\pm 30\%$) ranges; A2a re-plans every period and A2b never re-plans; A3 plans with normal travel times and ignores landslide closures until a dispatch fails.

Table 3 Compared methods

Code	Method	Uncertainty handling	Re-planning
B1	Static deterministic allocation	Point forecast	Once
B2	Proportional-to-risk heuristic	None	Every period
B3	Two-stage stochastic programming	Sampled scenarios	Every period
B4	Rolling budgeted robust optimization	Hand-tuned fixed budget	Every period
B5	Learned score-based dispatcher	Learned in simulator	Every period
FU	Forecast-based allocation	Uncalibrated point forecast	Every period
FC	Calibration-guided allocation	Calibrated point forecast	Every period
CGRR	Proposed (Algorithm 1)	Conformal ranges	Triggered

6.9 Sensitivity and stress scenarios

We vary $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$ and $\eta \in \{0, 0.05, 0.10, 0.20, 0.30\}$, and test four stress scenarios: S1, realized demand 30% higher than the exposure model implies; S2, 25% fewer resources; S3, forecast errors doubled at test time without recalibration, so that assumption A1 fails; S4, a sudden escalation in which demand doubles from mid-event in the three most exposed units, unforeseen by the forecasts. Because realized demand is a modelled proxy rather than observed emergency demand, results are also recomputed for mean need shares of 1%, 2% and 5% of the exposed population.

7 Evaluation metrics

Let $s_{i,t} = (d_{i,t} - \mathrm{cov}_{i,t})^+$ be the realized shortfall of unit i in period t, $D = \sum_{i,t} d_{i,t}$ the total realized demand and $\mathrm{RT}_{i,t}$ the time to the first passable resource arrival, capped at $\tau^{\max}$ for unserved demand. All metrics are computed on realized, not forecast, demand.

7.1 Service metrics

$$\text{Total weighted unmet demand:} \quad \mathrm{TWU} = \textstyle\sum_{i,t} w_i \, s_{i,t}, \tag{13}$$

$$\text{Worst-unit weighted unmet demand:} \quad \mathrm{WWU} = \max_i \textstyle\sum_t w_i \, s_{i,t}, \tag{14}$$

$$\text{Population coverage:} \quad \mathrm{COV} = 1 - \textstyle\sum_{i,t} s_{i,t}/D, \tag{15}$$

$$\text{Mean response time:} \quad \overline{\mathrm{RT}} = \textstyle\sum_{i,t} d_{i,t}\,\mathrm{RT}_{i,t}/D, \tag{16}$$

with the demand-weighted 95th percentile RT_{95} and the share of demand reached within 6 h reported alongside (16). Missed high-risk unit-periods count the (i,t) whose true class is severe and where $s_{i,t} > 0.5\,d_{i,t}$.

7.2 Operational metrics

Team relocations $\mathrm{REL} = \sum_t \sum_{r,j,i} y^{+}_{r,j,i,t}$ over the event; the number of re-plans; failed dispatches, i.e. resources sent on a route that was impassable at the time of travel because of flooding or a landslide; and computation time per re-plan.

7.3 Forecast and demand metrics

Forecast skill on the observed classes is reported as accuracy, macro-averaged precision, recall and F1, and macro one-versus-rest ROC-AUC. Calibration is reported as the top-label expected calibration error (ECE, 10 bins), the multi-class Brier score and the NLL, before and after temperature scaling, with reliability diagrams. Demand estimation is assessed by the mean absolute error (MAE) and root mean square error (RMSE) of $\hat{d}_{i,t}$ against the realized demand proxy, and by the rank correlation with relief-camp occupancy.

7.4 Fairness and utilization

Fairness is the Gini coefficient of the event-level share of demand served across units ($0 =$ equal). Utilization is the share of dispatched capacity $\sum \kappa_r x$ that served demand. Uncovered exposed population counts exposed people in unit-periods where less than half of demand was served.

7.5 Reliability metrics

Empirical coverage of the conformal sets, $\frac{1}{|I|T} \sum_{i,t} \mathbf{1}\{k^{\star}_{i,t} \in C_{i,t}\}$, compared with the target $1-\alpha$; mean set size; and the exceedance rate $\frac{1}{|I|T} \sum_{i,t} \mathbf{1}\{s_{i,t} > a_{i,t}\}$, compared with the bound α of Proposition 2.

7.6 Statistical testing

For each metric, CGRR is compared with each baseline across the paired simulation runs with the Wilcoxon signed-rank test, with Holm correction for multiple comparisons. Results are reported as medians with interquartile ranges and effect sizes, not as single best runs.

8 Result evaluation

8.1 Forecast skill and calibration

The rainfall-based forecaster is weak but informative (Table 4): accuracy is 0.609 in 2018 and 0.463 in 2019 against 0.25 for a uniform guess, with macro-F1 of 0.298 and 0.234 and ROC-AUC of 0.675 and 0.701. Temperature scaling ($T = 1.425$) reduced the ECE on the calibration periods from 0.161 to 0.119 and on the 2019 event from 0.169 to 0.151, but increased it on the 2018 event from 0.107 to 0.150 (Table 5). The reliability diagrams explain the difference (Figs. 3a and 4a). On the calibration periods the forecasts were over-confident, so $T > 1$ softened them; during the 2018 flood the observed frequency of the predicted class exceeded the stated confidence in most bins, so softening moved the forecasts further from the diagonal. A calibration fitted on ordinary monsoon periods therefore did not transfer fully to the extreme event. Demand-estimation errors are of the order of a few hundred people per district and period (MAE 273 in 2018, 356 in 2019), and calibration changed them by less than 5%.

Table 4 Forecasting skill on the test events (macro-averaged precision, recall, F1 and one-versus-rest ROC-AUC over the four risk classes; temperature scaling does not change the predicted class)

Event	Accuracy	Precision	Recall	F1	ROC-AUC
2018	0.609	0.310	0.361	0.298	0.675
2019	0.463	0.259	0.283	0.234	0.701

Table 5 Forecast calibration before and after temperature scaling ($T = 1.425$, fitted on the calibration periods, where ECE fell from 0.161 to 0.119), and demand-estimation error in people per district and period

Event	Forecast	ECE	Brier	NLL	Demand MAE	Demand RMSE
2018	Uncalibrated	0.107	0.470	0.774	273	454
2018	Temperature-scaled	0.150	0.486	0.827	281	445
2019	Uncalibrated	0.169	0.560	0.938	356	587
2019	Temperature-scaled	0.151	0.556	0.956	369	574

8.2 Conformal coverage on the test events

At $\alpha = 0.10$ the conformal sets contained the observed class in 99.0% of district-periods in 2018 and 97.5% in 2019, above the 90% target, with mean set sizes of 2.34 and 2.42 of the four classes. Coverage rose monotonically as α fell, from 0.78 at $\alpha = 0.30$ to 0.995 at $\alpha = 0.05$ in 2018 (Fig. 3b), and exceeded its target at every level in both events (in 2019, from 0.708 at $\alpha = 0.30$ to 0.982 at $\alpha = 0.05$). The sets are conservative because the test-event forecasts were less confident than those of the calibration periods. The exceedance rate, the share of district-periods whose realized shortfall exceeded the planned upper shortfall $a_{i,t}$, was 0.153 in 2018 and 0.250 in

2019, above $\alpha = 0.10$. The bound of Proposition 2 is therefore not met in the hindcast, because its assumption (A2) does not hold: realized demand is drawn from a need share that does not depend on the risk class, so it can exceed $v_i P_i \delta_{k^\star}$ even when the class is covered. Set coverage is guaranteed by the conformal construction; demand coverage additionally requires the damage fractions to bound realized need.

8.3 Comparison with baselines

On the 2018 hindcast CGRR covered 76.5% of modelled demand (interquartile range 66.1–87.4%), significantly more than the static, proportional, fixed-budget robust, learned and forecast-only policies (36.3–65.6%; Table 6), with lower total and worst-district unmet demand than each of them. Only two-stage stochastic programming (B3) did better on service: it covered 85.2% (73.9–94.2%) and left a median of 31,934 fewer people unserved per run. B3 achieved this by re-solving at every one of the 96 periods; it made 190 team relocations against 109 for CGRR (CGRR re-planned 20 times), had five failed dispatches against one, and was slower to reach people (mean response 0.96 against 0.78 h, 95th percentile 2.08 against 1.48 h). Both methods missed a single high-risk district-period. B3 also served demand more evenly across districts (Gini 0.031 against 0.064), and CGRR was second on fairness among all methods. The learned dispatcher B5 relocated teams 5,547 times, showing why every-period re-planning without a relocation cost is impractical. All differences between CGRR and each baseline in Table 6 are significant after Holm correction, except the P95 response time against B1, B4, FU and FC, which is identical.

Table 6 Allocation performance on the 2018 Kerala hindcast: median over 200 runs. Bold: best; *: CGRR differs significantly from the method (Wilcoxon signed-rank, Holm-corrected $p < 0.05$). Interquartile ranges are given in Online Resource 1

Method	Worst-unit weighted unmet	Total unmet (people)	Mean response (h)	P95 response (h)	Coverage (%)	Missed high-risk	Reloca-tions
B1 Static	138,870*	281,293*	0.78*	**1.48**	36.3*	5.0*	**0***
B2 Proportional	82,750*	159,526*	1.11*	2.75*	63.0*	3.0*	997*
B3 Stochastic	**53,543***	**69,147***	0.96*	2.08*	**85.2***	**1.0***	190*
B4 Fixed robust	101,830*	152,695*	**0.78***	**1.48**	65.6*	2.0*	77*
B5 Learned	82,481*	167,604*	1.13*	2.75*	62.7*	3.0*	5,547*
FU Uncalibrated	116,613*	192,859*	**0.78***	**1.48**	55.6*	3.5*	43*
FC Calibrated	114,526*	182,789*	**0.78***	**1.48**	57.7*	3.0*	63*
CGRR	72,937	101,557	0.78	**1.48**	76.5	**1.0**	109

8.4 Independent test: the 2019 Kerala floods

The 2018 ranking does not hold on the held-out 2019 event (Table 7). With nine teams, 18 boats and eight medical units against a modelled need of the order of 4×10^5 people, every method covered between 27.4% and 29.5% of demand, and CGRR's coverage did not differ significantly from that of any baseline except B1 (static), which covered 2.0 points more. CGRR had the lowest worst-district unmet demand (117,669 against 122,991–125,563 for the baselines; all differences significant) and re-planned in

69 rather than 84 periods with 391 relocations, fewer than every re-planning baseline except FU. Its 95th-percentile response time reached the 12-h cap, as did that of B4: both robust methods concentrated the few resources on the districts with the largest upper demand and left others unserved. When resources fall this far short of need, no allocation method can change aggregate coverage; the choice of method mainly decides which districts go unserved and how often teams move.

Table 7 Allocation performance on the 2019 Kerala hindcast (independent test): median over 200 runs; notation as in Table 6

Method	Worst-unit weighted unmet	Total unmet (people)	Mean response (h)	P95 response (h)	Coverage (%)	Missed high-risk	Reloca-tions
B1 Static	123,695*	**293,336***	**0.94***	2.08*	**29.5***	4.0	**0***
B2 Proportional	122,991*	301,030	1.28*	3.72*	27.8	4.0*	1,365*
B3 Stochastic	125,563*	299,584	1.14*	2.08*	27.6	4.0*	735*
B4 Fixed robust	125,132*	299,034	2.07*	12.00*	28.2	4.0	579*
B5 Learned	122,991*	301,110	1.28*	3.72*	27.8	4.0*	1,364*
FU Uncalibrated	124,541*	301,700	1.23*	**1.73***	27.4	4.0	349*
FC Calibrated	123,682*	299,309	1.17*	2.90*	28.0	**3.0**	657*
CGRR	**117,669**	305,489	1.44	12.00	28.1	**3.0**	391

8.5 RQ1: value of calibration and calibrated demand ranges

Calibration alone added little: FC covered 57.7% of 2018 demand against 55.6% for FU, and its worst-district unmet demand was 2% lower. The large gain came from the calibrated ranges: CGRR covered 18.8 points more than FC and reduced worst-district unmet demand by 36%. Ablation A1, which keeps triggered re-planning but replaces the conformal ranges with fixed ±30% ranges, covered 65.4%, so the conformal sets account for 11.1 of those points and for a 28% reduction in worst-district unmet demand (Table 8). CGRR also left fewer exposed people in district-periods with less than half of demand served (2.5 million person-periods against 4.9–14.3 million for all baselines except B3, with 1.5 million) and had a lower Gini coefficient (0.064) than every method except B3. Its utilization, the share of dispatched capacity that met demand, was 35%: lower than FU, FC, B4 and B1, which dispatch fewer units, and higher than B2, B3 and B5.

Table 8 Ablation study on the 2018 hindcast (median over 200 runs)

Variant	Worst-unit weighted unmet	Coverage (%)	Relocations	Re-plans	Failed dispatches
CGRR (full)	72,937	76.5	109	20	1
A1 no conformal ranges	101,830	65.4	76	21	0
A2a re-plan every period	72,972	76.5	109	96	1
A2b never re-plan	104,305	56.9	0	1	24
A3 no hazard-aware routing	70,918	76.6	75	21	19
FC no robustness, no trigger	114,526	57.7	63	96	0

The reliability level α acts as a single setting for how cautious the plan is (Table 9). Lowering α from 0.10 to 0.05 raised 2018 coverage from 76.5% to 84.0%, within 1.2 points of B3, at the cost of 181 relocations (B3: 190) and 41 re-plans (B3: 96); raising it to 0.20 or 0.30 cut relocations to about 60 but coverage to about 60%. Compared with the hand-tuned budget of B4, which covered 65.6%, the calibrated ranges at the default $\alpha = 0.10$ gave 10.9 points more coverage and 28% less worst-district unmet demand with 32 more relocations. Because α was fixed before the experiments, the $\alpha = 0.05$ result is a sensitivity finding, not a tuned result.

Table 9 Sensitivity to the reliability level α at $\eta = 0.10$ (2018 and 2019, median over 200 runs): empirical set coverage, mean set size, demand coverage, relocations and re-plans

Event	α	Set coverage	Mean set size	Coverage (%)	Relocations	Re-plans
2018	0.05	0.995	2.73	84.0	181	41
2018	0.10	0.990	2.34	76.5	109	20
2018	0.20	0.906	1.80	59.9	55	8
2018	0.30	0.779	1.46	59.4	60	10
2019	0.05	0.982	2.91	30.2	242	66
2019	0.10	0.975	2.42	28.1	391	69
2019	0.20	0.836	1.79	26.4	301	60
2019	0.30	0.708	1.48	27.6	388	62

8.6 RQ2: triggered versus every-period re-planning

The trigger rule matched the coverage of re-planning at every period with a fifth of the re-plans. In 2018, A2a (re-plan every period) covered 76.5%, the same as CGRR, with 96 re-plans against 20 and identical relocations, while A2b (never re-plan) covered 57.0% and had 24 failed dispatches as roads closed under a plan fixed on the first day (Table 8). At $\alpha = 0.10$, raising η from 0 to 0.30 reduced the re-plans from 96 to 14 with coverage unchanged at 76.5%. Of the 19 re-plans after the initial plan, 18 were triggered by a risk-class escalation, four of these together with a route failure, and one by a forecast shift (Fig. 3c). They cluster at the onset of the flood on 8–11 August, at its peak on 16–18 August and in the late-August rains of 26–29 August. In 2019 the rule fired in 69 of 84 periods, mostly on forecast shifts (52) and escalations (45). With only nine teams, the capacity floor C in (12) is small, so modest changes in forecast demand exceed η; the threshold should be set relative to the event's capacity.

8.7 RQ3: inundation- and landslide-aware routing

Hazard-aware routing prevented almost all failed dispatches: ablation A3, which plans with normal travel times and ignores closures until a dispatch fails, sent resources on impassable routes 19 times in 2018 against once for CGRR, and its mean and 95th-percentile response times were longer (0.83 against 0.78 h and 1.67 against 1.48 h). Its effect on aggregate coverage was negligible (76.6% against 76.5%), because the failed dispatches were few relative to the resources deployed and were corrected at the next re-plan; A3 also made fewer relocations (75 against 109), since CGRR moves teams

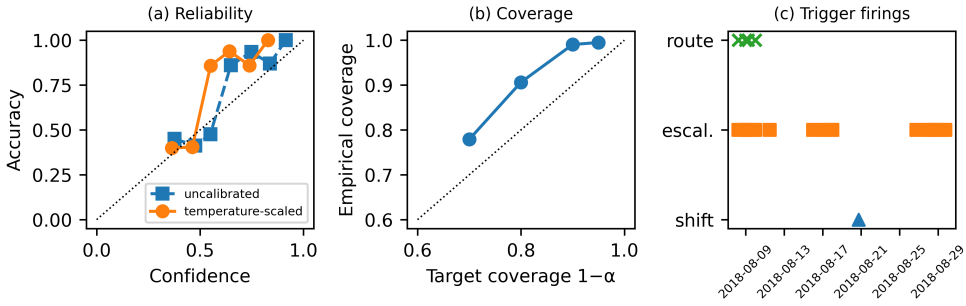


Fig. 3 2018 Kerala hindcast. (a) Reliability diagram of the forecasts before and after temperature scaling; (b) empirical coverage of the conformal sets against the target $1 - \alpha$; (c) periods in which the trigger rule re-planned, by cause (forecast shift, risk-class escalation, route failure)

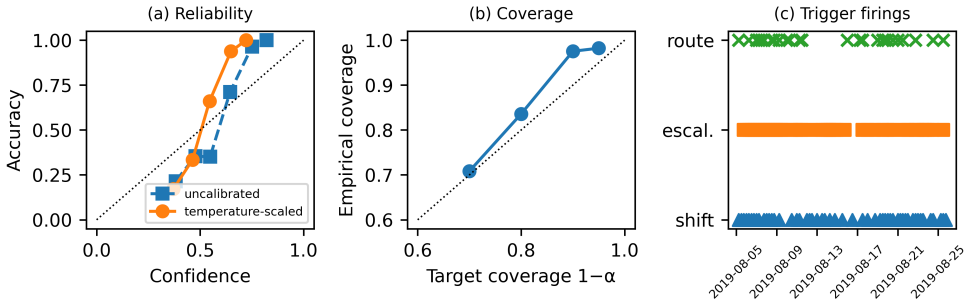


Fig. 4 2019 Kerala hindcast (independent test); panels as in Fig. 3

away from routes it expects to flood. The hindcast does not record whether each failure was caused by flooding or by a landslide. In 2019 neither landslide site blocked the main-road network, and no method had more than one failed dispatch, so RQ3 can be answered only for 2018.

8.8 Sensitivity and stress scenarios

Under every stress scenario the best baseline in 2018 remained B3 (Table 10). CGRR left 31% more people unserved than B3 when demand was 30% higher (S1), 19% more with a quarter fewer resources (S2) and 20% more under sudden escalation (S4), but reached people faster in all four scenarios (95th-percentile response 11–30% shorter). When forecast errors were doubled without recalibration (S3), the gap nearly closed (1.5% more unmet demand, 1.3 points less coverage), consistent with the conformal ranges absorbing part of the forecast error. In 2019 the best baseline was the static plan B1, and CGRR stayed within 3.1% of its unmet demand in every scenario, with the 95th-percentile response at the 12-h cap. The ranking of methods was stable across the assumed need share (Online Resource 1): in 2018 CGRR covered 93.9%, 76.5%

Table 10 Robustness: change of CGRR relative to the best baseline (highest coverage) under each stress scenario, median over 200 runs; positive unmet-demand and P95 changes and negative coverage changes favour the baseline

Scenario	Best baseline	Unmet demand change (%)	P95 response change (%)	Coverage change (pp)
2018				
S1 Severe flood (+30% demand)	B3	+31.1	-28.8	-8.2
S2 Resource shortage (−25%)	B3	+18.8	-11.2	-3.5
S3 Forecast degradation	B3	+1.5	-28.8	-1.3
S4 Sudden escalation (×2, 3 units)	B3	+19.9	-30.2	-6.7
2019				
S1 Severe flood (+30% demand)	B1	+1.8	+477.1	-0.9
S2 Resource shortage (−25%)	B1	+3.1	+477.1	-2.3
S3 Forecast degradation	B1	+1.9	+0.0	-2.0
S4 Sudden escalation (×2, 3 units)	B1	-0.3	+0.0	-0.3

and 46.0% of demand at mean need shares of 1%, 2% and 5%, second to B3 (98.9%, 85.2%, 54.8%) and ahead of all other methods; in 2019 all methods were within 4.2 points of each other at each share. Changing η altered the number of re-plans much more than coverage (Sect. 8, RQ2).

8.9 Discussion

8.9.1 Implications for disaster management authorities

The framework replaces a hand-tuned uncertainty budget with one setting, α, that authorities can read as a reliability target: plan so that, for at least $(1 - \alpha)$ of units, demand will not exceed what was prepared for. The trigger rule gives the Kerala State Disaster Management Authority, District Disaster Management Authorities and NDRF a stated reason for each redeployment, which a learned dispatcher cannot easily provide; such interpretability addresses the limited practitioner uptake of humanitarian optimization models (Rodríguez-Espíndola et al, 2023). Because routes depend on predicted inundation and mapped landslides, the plan avoids sending road-based teams along roads the forecast already expects to flood or that are blocked, and sends boats or alternative routes instead. The results also show the limits of the approach. In 2018 CGRR clearly outperformed allocation that ignores forecast uncertainty (FU, FC), allocation with a hand-tuned uncertainty budget (B4) and the learned dispatcher (B5), and it did so with a fifth of their re-plans. It did not match two-stage stochastic programming, which optimizes expected shortfall over sampled demand scenarios and re-solves every period; B3 covered 8.7 points more at the default α and 1.2 points more at $\alpha = 0.05$, at the cost of 96 re-plans, more relocations and slower responses. For an authority, the choice between them is a choice between maximum expected coverage and a plan that changes rarely, comes with a stated reason for each change and is tuned by one reliability setting. Combining the calibrated ranges with scenario-based recourse is a natural next step. In 2019, resources were so far below need that no method changed aggregate coverage; the framework's contribution there was lower worst-district unmet demand and fewer relocations.

8.9.2 Relation to the forecasting chain

The value of the allocation depends on forecast calibration. Model reliability can shift between periods (Xiao et al, 2026), so the calibration set for the conformal layer should be refreshed from recent events. Scenario S3 shows how the plan degrades when coverage is lower than promised. Because the severe class is rare, extreme conformal methods (Pasche et al, 2026) may give tighter guarantees where they matter most.

8.9.3 Limitations

Demand is a modelled proxy derived from exposure and an assumed need share, not observed emergency demand, and simulated operational scenarios are kept distinct from observations throughout; relief-camp occupancy is used as an independent check ($\rho = 0.60$) but records people who reached camps, not those still waiting for rescue. Because the need share does not depend on the risk class, assumption (A2) of Proposition 2 does not hold and the observed exceedance rates (0.15 and 0.25) exceed α. The forecasting layer is a simple rainfall-based model with modest skill and no depth output, so inundation-dependent travel times use a depth proxy; a stronger forecaster with depth would test the framework more fully. Sentinel-1 observes each district only on some dates, exposure before the first scene of each event is set to zero, and flooded roads are known only on imaged dates. The boat and medical stocks and their split across hubs are assumptions, and no district vulnerability index was used. Landslide closure durations are approximate where reopening dates are not recorded, and the two 2019 landslides did not block the main-road network. Proposition 2 also relies on exchangeability and marginal coverage, which serially dependent flood data satisfy only approximately. The model does not represent crew fatigue, communication failures or coordination between agencies, and is solved one period at a time without multi-period look-ahead. Solve times grow with the number of units and resource types, so finer spatial units may need decomposition or heuristics.

9 Conclusion and future work

This study proposed a calibration-guided robust reallocation framework that links the measured reliability of flood risk forecasts to the allocation of emergency resources, on a road network that degrades with flood depth and landslide blockage. On hindcasts of the 2018 and 2019 Kerala floods, five findings stand out. (1) Calibrated demand ranges improved allocation substantially (RQ1): in 2018 CGRR covered 76.5% of modelled demand against 57.7% for calibrated point forecasts and 65.6% for a hand-tuned robust budget, and the conformal ranges alone accounted for 11.1 points and a 28% reduction in worst-district unmet demand. (2) The trigger rule matched every-period re-planning at 76.5% coverage with 20 instead of 96 re-plans (RQ2). (3) Hazard-aware routing reduced failed dispatches from 19 to one and shortened response times, with little effect on aggregate coverage (RQ3). (4) Two-stage stochastic programming remained the strongest baseline, covering 85.2% at the cost of re-planning every period and 74% more relocations; lowering α to 0.05 brought CGRR to 84.0% with 41 re-plans. (5) Under the severe scarcity of 2019 all methods covered 27–30% of demand; CGRR had the lowest worst-district unmet demand and fewer relocations, but its 95th-percentile

response time reached the 12-h cap. Temperature scaling fitted on ordinary monsoon periods did not improve calibration during the extreme 2018 event, which argues for recalibrating on recent events.

For practice, we recommend that State and District Disaster Management Authorities (i) treat α as a planning reliability target agreed before the monsoon; (ii) refresh the calibration set every monsoon season; (iii) maintain a road-segment landslide record with closure and reopening dates, since route failures triggered a fifth of the redeployments in 2018 and more than a third in 2019; and (iv) log the trigger condition for each redeployment to support after-action review.

Future work will extend the framework to multi-hazard allocation in which landslides are forecast rather than only observed, add multi-period look-ahead, model coordination across agencies, and test transfer to other flood-prone states such as Assam and Bihar.

Appendix A Proof of Proposition 1

Fix the coverage cov and write $\Delta_i = d_i^U - \hat{d}_i \geq 0$, $u_i^0 = (\hat{d}_i - \mathrm{cov}_i)^+$, $a_i^0 = (d_i^U - \mathrm{cov}_i)^+$ and $g_i = w_i(a_i^0 - u_i^0) \geq 0$. The worst-case weighted unmet demand is

$$W = \max_{z \in Z} f(z), \qquad f(z) = \sum_i w_i \big(\hat{d}_i + z_i \Delta_i - \mathrm{cov}_i\big)^+,$$

with $Z = \{z : 0 \leq z \leq 1,\ \sum_i z_i \leq \Gamma\}$. Each term of f is convex in z_i, so f is convex and attains its maximum over the polytope Z at a vertex. Because the constraint matrix of Z is totally unimodular and Γ is an integer, every vertex is a 0–1 vector with at most Γ ones. At such a vertex with support S, $f = \sum_{i \notin S} w_i u_i^0 + \sum_{i \in S} w_i a_i^0$, hence

$$W = \sum_i w_i u_i^0 + \max_{z \in Z} \sum_i g_i z_i = \sum_i w_i u_i^0 + \min \Big\{ \Gamma\theta + \sum_i \pi_i : \theta + \pi_i \geq g_i,\ \theta, \pi \geq 0 \Big\},$$

where the second equality is linear programming duality.

Now take any (u, a, θ, π) feasible for (6)–(7), so $u_i \geq u_i^0$ and $a_i \geq a_i^0$. For any S with $|S| \leq \Gamma$, summing (7) over S and using $\theta \geq 0$ gives

$$\begin{aligned} \sum_i w_i u_i + \Gamma\theta + \sum_i \pi_i &\geq \sum_i w_i u_i + \sum_{i \in S} w_i (a_i - u_i) \\ &= \sum_{i \notin S} w_i u_i + \sum_{i \in S} w_i a_i \ \geq \ \sum_{i \notin S} w_i u_i^0 + \sum_{i \in S} w_i a_i^0. \end{aligned}$$

Maximizing the right-hand side over S shows that the objective is at least W. Choosing $u = u^0$, $a = a^0$ and the optimal dual (θ, π) attains W. Therefore the minimum equals W. □

References

Angelopoulos AN, Bates S (2023) Conformal prediction: a gentle introduction. Foundations and Trends in Machine Learning 16(4):494–591. https://doi.org/10.1561/2200000101

Attar A, Ghasemi P, Ozkul S, et al (2026) Adaptive disaster relief logistics under uncertainty: a hybrid robust–fuzzy two-stage mathematical model with inter-regional sharing. Journal of the Operational Research Society https://doi.org/10.1080/01605682.2026.2730524

Avishan F, Elyasi M, Yanıkoğlu İ, et al (2023) Humanitarian relief distribution problem: an adjustable robust optimization approach. Transportation Science 57(4):1096–1114. https://doi.org/10.1287/trsc.2023.1204

Ben-Tal A, El Ghaoui L, Nemirovski A (2009) Robust Optimization. Princeton University Press, Princeton

Ben-Tal A, Chung BD, Mandala SR, et al (2011) Robust optimization for emergency logistics planning: risk mitigation in humanitarian relief supply chains. Transportation Research Part B: Methodological 45(8):1177–1189

Bertsimas D, Sim M (2004) The price of robustness. Operations Research 52(1):35–53. https://doi.org/10.1287/opre.1030.0065

Caunhye AM, Nie X, Pokharel S (2012) Optimization models in emergency logistics: a literature review. Socio-Economic Planning Sciences 46(1):4–13. https://doi.org/10.1016/j.seps.2011.04.004

Chothodi S, Kuniyil S, Patidar H, et al (2024) Linkages and reactions of geomorphic processes in Kerala flood, 2018. Natural Hazards 120(7):5953–5986. https://doi.org/10.1007/s11069-024-06469-5

Duan Y, Ding M, He Y, et al (2025) Projections of global road risk exposed to landslides under climate change. Communications Earth & Environment https://doi.org/10.1038/s43247-025-03073-8

Government of Kerala (2018) Additional memorandum: Kerala floods 2018, 1st august to 30th august 2018. Kerala State Disaster Management Authority. https://sdma.kerala.gov.in/wp-content/uploads/2019/08/Memorandum2-Floods-2018.pdf, accessed 30 October 2026

Government of Kerala and United Nations (2018) Kerala post disaster needs assessment: floods and landslides, August 2018. United Nations Development Programme. https://www.undp.org/publications/post-disaster-needs-assessment-kerala, accessed 27 October 2026

Guo C, Pleiss G, Sun Y, et al (2017) On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning, pp 1321–1330

Hao L, Rajaneesh A, van Westen C, et al (2020) Constructing a complete landslide inventory dataset for the 2018 monsoon disaster in Kerala, India, for land use change analysis. Earth System Science Data 12:2899. https://doi.org/10.5194/essd-12-2899-2020

He K, Carhart N, Pregnolato M, et al (2026) Probabilistic functionality assessment of road networks for medical emergency vehicles during flooding. Natural Hazards 122(6):233. https://doi.org/10.1007/s11069-026-08005-z

Huangfu Q, Hall JAJ (2018) Parallelizing the dual revised simplex method. Mathematical Programming Computation 10(1):119–150. https://doi.org/10.1007/s12532-017-0130-5

Huffman GJ, Stocker EF, Bolvin DT, et al (2023) GPM IMERG final precipitation L3 half hourly 0.1 degree x 0.1 degree V07. Goddard Earth Sciences Data and Information Services Center (GES DISC), Greenbelt, MD, https://doi.org/10.5067/GPM/IMERG/3B-HH/07

Inyang S, Taghikhah FR (2025) Digital twin-based decision support systems for natural disaster management: a systematic review of current trends and approaches. Science Talks 13:100406. https://doi.org/10.1016/j.sctalk.2024.100406

Jain J, Muñoz-Arriola F, Khare D (2025) Evidencing sources of flood disaster policy improvement leveraging flood risk attributes in India. Natural Hazards 121(7):8011–8038. https://doi.org/10.1007/s11069-025-07108-3

Johnstone C, Cox B (2021) Conformal uncertainty sets for robust optimization. In: Proceedings of the Tenth Symposium on Conformal and Probabilistic Prediction and Applications, pp 72–90

Liu H, Liang K, Yang J, et al (2026) Digital twin-driven dynamic optimization for urban emergency distribution. Industrial Engineering Journal 29(1):61–74. https://doi.org/10.3969/j.issn.1007-7375.250210

Manopiniwes W, Irohara T (2021) Optimization model for temporary depot problem in flood disaster response. Natural Hazards 105(2):1743–1763. https://doi.org/10.1007/s11069-020-04374-1

Metodiev D, Mendes RM, de Andrade MRM, et al (2026) Exploring rainfall thresholds for shallow landslides using effective half-life precipitation indices and cluster analysis: a case study in Santos and Guarujá, Brazil. Natural Hazards 122:697. https://doi.org/10.1007/s11069-026-08475-1

National Disaster Response Force (2018a) Kerala floods – 2018. https://ndrf.gov.in/en/node/89258, accessed 27 October 2026

National Disaster Response Force (2018b) Kerala floods: NDRF scaled up its rescue and evacuation operation, situation report of 18 August 2018. https://ndrf.gov.in/en/node/89184, accessed 30 October 2026

National Disaster Response Force (2019) Landslide in Kerala – 2019. https://ndrf.gov.in/en/node/89425, accessed 30 October 2026

OpenStreetMap contributors (2026) OpenStreetMap. https://www.openstreetmap.org, data extracted with OSMnx; accessed November 2026

Pasche OC, Lam H, Engelke S (2026) Extreme conformal prediction: reliable intervals for high-impact events. Extremes https://doi.org/10.1007/s10687-026-00536-9

Postance B, Hillier J, Dijkstra T, et al (2017) Extending natural hazard impacts: an assessment of landslide disruptions on a national road transportation network. Environmental Research Letters 12(1):014010. https://doi.org/10.1088/1748-9326/aa5555

Pregnolato M, Ford A, Wilkinson SM, et al (2017) The impact of flooding on road transport: a depth-disruption function. Transportation Research Part D: Transport and Environment 55:67–81. https://doi.org/10.1016/j.trd.2017.06.020

Prihatmanto AS, Prasetyadi A, Yoganingrum A, et al (2025) The digital twin city in enhancing flood evacuation systems: future opportunities and challenges. IEEE Access 13:32371–32383. https://doi.org/10.1109/ACCESS.2025.3539346

Rachit, Tripathi V, Mohanty MP, et al (2026) Rapid flood susceptibility mapping in the Indian Himalayan region using CNN-U-Net segmentation: insights from the 2025 monsoon events. Natural Hazards 122(10):424. https://doi.org/10.1007/s11069-026-08175-w

Rao GV, Nagireddy NR, Keesara VR, et al (2024) Real-time flood forecasting using an integrated hydrologic and hydraulic model for the Vamsadhara and Nagavali basins, Eastern India. Natural Hazards 120(7):6011–6039. https://doi.org/10.1007/s11069-023-06366-3

Rodríguez-Espíndola O, Ahmadi H, Gastélum-Chavira D, et al (2023) Humanitarian logistics optimization models: an investigation of decision-maker involvement and directions to promote implementation. Socio-Economic Planning Sciences 89:101669. https://doi.org/10.1016/j.seps.2023.101669

Saharia M, Jain SK, Prakash V, et al (2025) A district-level flood severity index for flood management in India. Natural Hazards 121(15):17815–17831. https://doi.org/10.1007/s11069-025-07493-9

Shen Y, Wang X, Wang H, et al (2023) A dynamic task assignment model for aviation emergency rescue based on multi-agent reinforcement learning. Journal of Safety Science and Resilience 4(3). https://doi.org/10.1016/j.jnlssr.2023.06.001

Tiwari V, Kumar V, Matin MA, et al (2020) Flood inundation mapping – Kerala 2018; harnessing the power of SAR, automatic threshold detection method and Google Earth Engine. PLOS ONE 15(8):e0237324. https://doi.org/10.1371/journal.pone.0237324

Vega J, Marín-Rodríguez NJ, Hurtado MI (2026) Landslides impacting road infrastructure under climate and weather influences: a systematic literature review with descriptive bibliometric insights. Natural Hazards 122:699. https://doi.org/10.1007/s11069-026-08467-1

Vovk V, Gammerman A, Shafer G (2005) Algorithmic Learning in a Random World. Springer, New York

WorldPop (2018) Global high resolution population denominators project. School of Geography and Environmental Science, University of Southampton, https://doi.org/10.5258/SOTON/WP00645

Xiao T, Segoni S, Liang X, et al (2026) Same area, different timeframes: unveiling the evolution of landslide patterns and their impact on susceptibility maps. Natural Hazards 122:678

Yan L, Mahmud S, Shen H, et al (2020) MobiRescue: reinforcement learning based rescue team dispatching in a flooding disaster. In: Proceedings of the IEEE International Conference on Distributed Computing Systems (ICDCS)

Zhang J, Lu M, Zhang L, et al (2021) Assessing indirect economic losses of landslides along highways. Natural Hazards 106:2775–2796. https://doi.org/10.1007/s11069-021-04566-3

Zhang J, Long DZ, Li Y (2023) A reliable emergency logistics network for COVID-19 considering the uncertain time-varying demands. Transportation Research Part E: Logistics and Transportation Review 172:103087. https://doi.org/10.1016/j.tre.2023.103087

Zhang Q, Yu H, Li Z, et al (2020) Assessing potential likelihood and impacts of landslides on transportation network vulnerability. Transportation Research Part D: Transport and Environment 82:102304. https://doi.org/10.1016/j.trd.2020.102304

Zio E, Miqueles L (2024) Digital twins in safety analysis, risk assessment and emergency management. Reliability Engineering & System Safety 246:110040. https://doi.org/10.1016/j.ress.2024.110040

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- **Code availability:** The forecasting interface, calibration and conformal layers, optimization model, trigger rule, simulator, baselines, Colab notebooks and run logs are available at https://github.com/dmounikaresearch-cmyk/kerala-flood-cgrr (MIT licence), archived at Zenodo [DOI].
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